Sameer Gupta

Sameer Gupta

2026

Sameer Gupta

xameer@disroot.org | +91 799 311 0170 | Ananthapur, India.

Summary

Infrastructure architect and functional programmer with 15 years of applied experience across systems design, data engineering, and API development. I specialise in reproducible infrastructure as code, service design, DSL and language inter-ops, binary cache optimisation, and zero-trust networking. Track record of reducing cloud compute and hosting costs through self-hosted, declarative infrastructure. Available for remote senior and staff engineering roles.

I commit to git forges and write on my blog and microblog.

Core Competencies

Infrastructure as Code Terraform/Pulumi, GitLab CI/CD, declarative VM provisioning, system service design.

Programming Languages & Semantics NixPkgs, Haskell, Rust, Lua, Bash, Python, Wasm, Emacs Lisp.

Distributed Systems Kubernetes (bare-metal, CRI-O, etcd, secretbox), Apps and services for zero-trust network meshes, Cloudflare Workers.

Binary Caching and CI Optimisation Cachix, Attic (R2-backed), FastCDC chunk tuning, 80% CI time reduction through cache layer design.

Storage and Data MinIO (S3-compatible), Restic (offsite backup), Longhorn (K8S PVC), Turso/libSQL (SQLite), etcd, OpenEBS, AWS CSI.

API and Service Design Protocol implementation, CRUD and REST web-hooks.

Selected Projects

Self-Hosted Blog Infrastructure (2026): For a small-scale DevOps, with Rust and Haskell toolchain, but without the cloud or CI runner compute cost or CDN egress fees and reduced network costs.
NixOS Kubernetes Cluster (2025–26): A production-grade n-node QEMU VM cluster, with a reproducible cluster state
For Apps, I used GitLab webhook receivers triggering Kubernetes rolling restarts on successful pipeline events. Which are exposed via Cloudflare Tunnel.
Cardano Proposal (2023–24): Designed and wrote the winning proposal for an EDSL targeting Plutus smart contract development on open-source hardware addressing scalability bottlenecks and hardware compatibility constraints that limited Cardano sidechain deployment. Funded via community vote on Ideascale; partnered with D.A.O. Konma (Chennai) for pitch.

Experience

2024–25 | Freelance Trainer, Data and Cloud Infrastructure – Hyderabad

Trained engineers in data lake architecture: Medallion and Data Vault patterns, cloud-native Kubernetes, and hybrid cost/latency models. Covered Terraform IaC, Kubernetes resource management, GitOps, and disposable environment design.

2023–24 | Senior Haskell Developer – Haskledger

Developed a typed AST in Haskell, capturing smart-contract constructs, their low-level constraints, with a denotational interpreter mapping DSL expressions to Plutus Core via operational rewrite rules. Used GADTs and phantom types to make resource-exceeding contracts are unrepresentable at compile time, and a deep embedding layer for optimisation passes batched state transitions, compact validator scripts that reduce sidechain execution costs before evaluation.

2019–23 | Freelance Network Programmer

Prototyped interconnection services for crypto mining data centres. Full cycle: value proposition, feasibility, simulation, prototype, pitch. I developed Linux services with Podman, cloud, and serverless on tunnels.

2016–19 | Freelance Web Developer

Frontend features and UI fixes for seed-funded startups. Web administration, development, and App firewalls.

2015–16 | Data Science Lead – Examify, Mumbai

NLP pattern recognition, ML with CUDA. Reported to CTO and CEO.

2010–12 | Faculty, Physics

Preparatory physics for IIT-JEE at commercial institutes.

Education

2006–2010 | B.Tech, M.E.M.S. – Indian Institute of Technology Bombay. I was enrolled on the 4-year course. Applied Physics, Mathematics, CS-101. Winter intern at I.I.M. Bangalore (traffic policy research and analysis).

Dorm startup *JeeCarnot*: statistical analysis of competitive exam outcomes.

Open Source

- A serverless API for a virtual Tailscale microservice, deployed on the cluster and natively as a systemd service to post comments on my blog, and comment data stored with Turso libsql Connector. So that anyone with a valid GPG key or OpenID can comment on my blog post, even from the machines that are not my Tailscale peer.
- A fast CLI tool to crawl RSS/Atom job feeds from an OPML file, deduplicate entries, and output recent jobs in multiple formats.

Communication

Primary: Hindi
Secondary: English